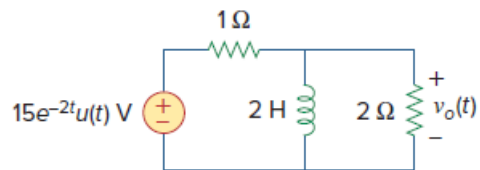


Problem

Find $v_o(t)$ in the circuit shown in Fig. 16.9. Note that, since the voltage input is multiplied by $u(t)$, the voltage source is a short for all $t < 0$ and $iL(0) = 0$.

Figure 16.9:



Step-by-step solution

Step 1 of 4

Refer to the circuit in Figure 16.6 in the textbook.

Transform the circuit into s-domain.

$$15e^{-2t}u(t) \Rightarrow \frac{15}{s+2}$$

$$2\text{ H} \Rightarrow sL = 2s$$

Draw the modified circuit as shown in Figure 1.

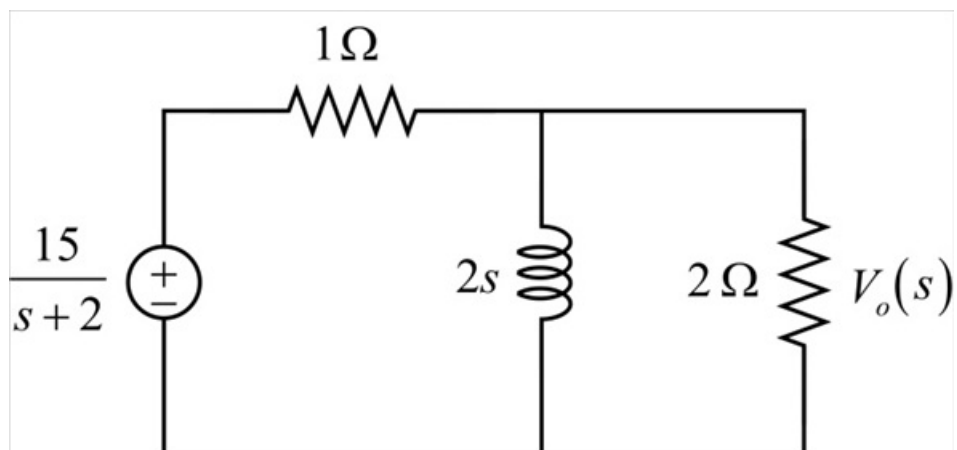


Figure 1

Step 2 of 4

Apply nodal analysis at node $V_o(s)$.

$$\frac{V_o(s) - \frac{15}{s+2}}{1} + \frac{V_o(s)}{2s} + \frac{V_o(s)}{2} = 0$$

$$V_o(s) \left[1 + \frac{1}{2s} + \frac{1}{2} \right] = \frac{15}{s+2}$$

$$V_o(s) \left[\frac{2s+1+s}{2s} \right] = \frac{15}{s+2}$$

$$V_o(s) = \frac{30s}{(s+2)(3s+1)}$$

$$V_o(s) = \frac{10s}{(s+2) \left(s + \frac{1}{3} \right)}$$

Step 3 of 4

By partial fractions,

$$V_o(s) = \frac{A}{s+2} + \frac{B}{s + \frac{1}{3}} \dots\dots (1)$$

$$10s = A \left(s + \frac{1}{3} \right) + B(s+2)$$

Comparing the coefficients,

$$10 = A + B$$

$$\frac{A}{3} + 2B = 0$$

By solving the two equations,

$$A = 12$$

$$B = -2$$

Step 4 of 4

Substitute 12 for A and -2 for B in equation (1).

$$V_o(s) = \frac{12}{s+2} - \frac{2}{s + \frac{1}{3}}$$

Apply inverse Laplace transform.

$$\begin{aligned} v_o(t) &= 12e^{-2t}u(t) - 2e^{-\frac{1}{3}t}u(t) \\ &= (12e^{-2t} - 2e^{-t/3})u(t) \text{ V} \end{aligned}$$

Thus, the voltage $v_o(t)$ is $\boxed{(12e^{-2t} - 2e^{-t/3})u(t) \text{ V}}$.